Few-Shot Hyperspectral Image Classification Based on Prototype Mask Contrast and Cross-Stage Feature Refinement

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***Abstract*—Recently, cross-domain few-shot learning (FSL) has achieved remarkable performance in hyperspectral image classification (HSIC). However, current prototype-based FSL methods overlook the redundancy between prototypes, which may lead to intradomain redundancy dependence and interfere with the effective alignment of data distributions between the source domain (SD) and the target domain (TD). To address these issues, we propose a prototype mask contrast and cross-stage feature refinement (PMC-CFR) method for few-shot HSIC. Specifically, a neighbor-resistant multidimensional feature fusion (NRMFF) network is designed to fuse features inclined toward different dimensional information after random neighboring-pixel removal. Next, the parameter-free adaptive mask generator (PAMG) identifies potential redundant components based on the statistical distribution of intradomain prototypes to form negative pairs, while treating other components as key features to form positive pairs. Contrastive pairs are further processed through graph convolution (GC) for supervised contrastive learning (SCL), which mitigates the dependence on intradomain redundancy and enhances the learning of key features. To extract more representative features, multilevel representations are formed depending on whether the shared multilayer perceptron (MLP) is applied to focus on key components in each domain by fitting positive pairs. Meanwhile, the directional calibration (DC) separates interclass prototypes across levels by balancing direction and distance. The synergy between MLP and DC in hierarchical feature modeling and semantic separation enables cross-stage feature refinement. Experimental results on four public HSI datasets demonstrate that the PMC-CFR outperforms many advanced contrastive methods, and feature visualizations further support its effectiveness in cross-domain feature distribution alignment.**

***Index Terms*—Cross-domain few-shot learning (FSL), feature refinement, graph convolution (GC), hyperspectral image classification (HSIC), supervised contrastive learning (SCL).**

Received 31 March 2025; revised 20 May 2025 and 9 July 2025; accepted 22 July 2025. Date of publication 24 July 2025; date of current version 5 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62406237, Grant 62176200, and Grant 12326617; in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2021A1515110686; in part by the Fundamental Research Funds for the Central Universities under Grant ZYTS25211; in part by the Postdoctoral Research Project Funding of Shaanxi under Grant 2023BSHTBZZ30; in part by the Project funded by China Postdoctoral Science Foundation under Grant 2024M752536; and in part by the Natural Science Basic Research Program of Shaanxi under Grant 2022JC-45. *(Corresponding author: Weitong Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2025.3592406

I. Introduction

HYPERSPECTRAL image (HSI) provides abundant spatial and spectral details by capturing spectral data across numerous narrow bands [1]. Leveraging this characteristic, the HSI classification (HSIC) technique can accurately identify different ground materials and has been widely applied in environmental monitoring, mineral detection, precision agriculture, and other fields [2], [3], [4], [5], [6].

Early traditional methods based on statistics and machine learning rely on automated feature dimensionality reduction techniques, such as principal component analysis (PCA) [7], or manually designed feature engineering approaches, including morphological method [8], coupled with simple classifiers like the k-nearest neighbor (KNN) [9], and support vector machine (SVM) [10]. However, their limitations in modeling nonlinear features and capturing complex spatial–spectral relationships hinder their performance in practical scenarios. Recently, numerous well-crafted network architectures based on deep learning have emerged in the field of HSIC [11]. Li et al. [12] designed a 3-D convolutional neural network (3DCNN) to jointly extract spatial–spectral features. Xu et al. [13] employed a mask matrix to ensure training sample sparsity, while Shen et al. [14] integrated long-range contextual information for each pixel. Liu et al. [15] combined convolution with graph convolution (GC) to merge features at both pixel and superpixel levels. Xu et al. [16] enhanced convolution through a self-attention mechanism to integrate global and local multiscale features. Roy et al. [17] combined morphological convolution with self-attention to improve token interactions. Despite the impressive performance of these methods, training deep networks generally requires extensive labeled samples. However, due to the difficulties in data collection, the high demand for domain expertise, and time costs,

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the availability of labeled samples in practical HSI training is extremely limited. Thus, training deep networks with strong generalization under limited labeled samples remains a critical research challenge.

Drawing from the human ability to learn with limited examples, few-shot learning (FSL), a subfield of meta-learning, has been effectively utilized in HSIC [18]. Meta-learning focuses on learning how to learn [19]. The model employs episodic training in FSL to simulate few-shot tasks in the source domain (SD) with sufficient labels, enabling quick adaptation to new tasks in the target domain (TD) with limited labels. Most FSL-based methods adopt prototypical networks (PNs) [20], which represent each class by a prototype and make predictions based on prototype similarities. Liu et al. [21] classified the TD using a metric space learned from the SD, while Gao et al. [22] fine-tuned the source-trained model with a few labeled target samples. Although effective, early FSL methods often assume identical data distributions between the SD and TD, overlooking the domain shift in real-world scenarios. To further improve, Li et al. [23] proposed a deep cross-domain FSL (DCFSL) method, which combines meta-learning with domain adaptation. Xi et al. [24] designed the FSL with the class-covariance metric (CMFSL) method, relying on task-adaptive class covariance estimation for meta-task classification. Cheng et al. [25] leveraged causal learning to capture the causal relationships between data and classes while eliminating their noncausal associations. Ye et al. [26] proposed an adaptive domain-adversarial FSL (ADAFSL) method, which dynamically weights adversarial loss based on interclass alignment for global conditional distribution alignment. Hu et al. [27] introduced a dual-adjustment mode-based cross-domain meta-learning (DMCM) method, adopting intradomain calibration and cross-domain alignment to mitigate domain shift. Qin et al. [28] decoupled features into domain-common and domain-specific components to suppress knowledge biased toward the SD and facilitate specific information learning from the TD. Liu et al. [29] designed a hybrid multitask learning (HyMuT) method, merging self-supervised strategies with domain adaptation to learn shared representations from three auxiliary tasks. Feng et al. [30] extracted TD discriminative features with reduced SD bias through knowledge distillation and decoupled logit distillation to balance knowledge interaction.

The distribution bias, not fully eliminated by cross-domain learning strategies, may still introduce disruptive transfer knowledge into aligned TD features. Unsupervised contrastive learning (CL) [31] leverages sample properties to distinguish finer differences, further mitigating unnecessary alignment bias. Sohn [32] improved the contrastive metric by considering the identification of a positive sample among multiple negative samples. Chen et al. [33] found that mapping features specifically to a latent space for applying contrastive loss can enhance feature quality. Hang et al. [34] paired patches cropped from different modalities at the same spatial location for alignment. Ongoing progress in CL has further enhanced the task orientation of supervised CL (SCL). Zhang et al. [35] enhanced feature discriminability during CL by utilizing label information. Liu et al. [36] applied SCL to enforce internal constraints on prototypes for refinement. Li et al. [37] aligned label text prototypes with image prototypes across modalities, and leveraged neighborhood pixel mask-based SCL to learn semantically strong and mask-robust features. However, current implementations of CL mainly rely on local features extracted by traditional convolution, limiting the perception of global information. GC integrates local information and global structure by leveraging adjacency relationships [38], and it aggregates neighborhood information via interlayer propagation to capture higher-order features of nodes [39]. Zhang et al. [40] proposed the graph information aggregation cross-domain FSL (Gia-CFSL) method, employing a graph distribution information propagation mechanism to perform graph alignment and enhance cross-domain information interaction. Ye et al. [41] introduced a cross-domain FSL method based on GC contrast (GCC-FSL), applying GC to cross-domain prototype pairs matched based on Euclidean distance to implement SCL. Furthermore, when restricted to cross-scene with access only to the SD, Qin et al. [42] considered generating more reliable synthetic domains from the SD to extract domain-invariant features for direct generalization to the TD. When limited to the TD only, Khotimah et al. [43] first performed self-supervised pretraining on spatiospectrally masked patches, then fine-tuned via hybrid episode-based global–local prototype fusion to enhance feature stability.

Despite promising results, current FSL methods still have limitations. The impact of feature redundancy in HSIC tasks cannot be overlooked. Particularly in PN, prototype features are directly used to represent classes and may contain not only discriminative features but also certain redundant components. Effectively identifying and suppressing redundant information within prototypes is a key challenge, yet it is rarely considered. If there exist redundant features that exhibit commonality among prototypes within a single domain, prototypes from different classes may contain similar yet nondiscriminative features. On the one hand, such redundant components in single-domain meta-learning contribute minimally to class discrimination, excessive learning may lead the model to rely on redundancy for classification to some extent, resulting in intradomain redundancy dependence. On the other hand, redundant features in the SD and TD often exhibit distributional inconsistency. Aligning such features directly not only fails to benefit cross-domain generalization but may also cause the model to focus on class-irrelevant features, thereby weakening the alignment of discriminative features and further exacerbating domain shift. The distributional inconsistency of redundant features across domains may mislead the model's feature alignment direction. This article refers to the phenomenon of performance degradation caused by such an alignment shift as potential redundant negative transfer. Additionally, a model typically utilizes neighboring pixels to provide local contextual information for feature extraction. However, in FSL, it tends to rely on the neighborhood patterns of extremely limited training samples. When the neighboring pixels of a given class undergo significant changes during testing, the model may struggle to adapt to such

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neighborhood variations, thereby affecting its discriminative ability.

To address the above issues, this article proposes a prototype mask contrast and cross-stage feature refinement (PMC-CFR) method for few-shot HSIC. First, a neighbor-resistant multidimensional feature fusion (NRMFF) network is designed, which reduces dependence on neighborhood patterns via random patch masking and fuses features with different tendencies in spatial and spectral dimensions. Next, the parameter-free adaptive mask generator (PAMG) leverages the statistical distribution of intradomain prototypes to differentiate the importance of features in class representation, thereby forming contrastive pairs. GC further extracts high-order dependencies among these pairs to achieve SCL. The prototype mask contrast (PMC) brings class anchors closer to positives enriched with key features, while pushing them farther from negatives containing potential redundancy. Furthermore, the shared multilayer perceptron (MLP) is selectively applied to focus on key components in each domain, driven by positive pairs, to form multilevel features. Meanwhile, the directional calibration (DC) further refines interclass and intraclass features across levels under directional guidance. The shared MLP and DC work in coordination to achieve cross-stage feature refinement.

The contributions of this article are summarized as follows.

1) A PMC-CFR framework is proposed for HSIC, which promotes cross-domain feature alignment by effectively modeling and suppressing prototype redundancy, thereby achieving superior performance in cross-domain FSL.
2) PAMG is introduced to identify potential redundancy using intradomain prototype distributions. By capturing semantic differences among prototypes and leveraging GC to establish global contrastive relationships, the PMC mitigates overfitting due to intradomain redundancy dependence and enhances key feature learning.
3) To extract more representative features, the shared MLP focuses on domain-specific key features by fitting positive pairs to form a new level of representation, while the DC strengthens interclass separability and intraclass compactness across levels under directional guidance. The synergy between MLP and DC in hierarchical feature modeling and semantic separation enables comprehensive feature refinement in a cross-stage manner.

II. Methodology

Cross-domain FSL involves the SD dataset $\mathbf{D}_s$ and the TD dataset $\mathbf{D}_t$. $\mathbf{D}_s$ and $\mathbf{D}_t$ contain C_s and C_t classes, respectively, where C_s is greater than C_t. Each meta-task T includes a support set S and a query set Q. If S contains C classes with K samples per class, T is termed a C-way K-shot task, also known as an episode. The process of constructing T is the same for both the SD and TD. Taking the TD meta-task T_t as an example, C classes are randomly selected, and for each class, K and N nonoverlapping samples are chosen to consturct $S_t = \{(x_t^i, y_t^i)\}_{i=1}^{C\times K}$ and $Q_t = \{(x_t^j, y_t^j)\}_{j=1}^{C\times N}$, respectively. Here, y_t^i is the label of sample x_t^i. The labeled samples in $\mathbf{D}_t$ are insufficient to generate T_t. Therefore, the training samples in $\mathbf{D}_t$ can be augmented by adding random Gaussian noise [44], performing random flipping, or applying other data augmentation techniques. PMC-CFR performs meta-task training alternately on the SD and TD. This section uses the TD as an example for explanation.

TABLE I
Key Notations Used in PMC-CFR

Notation	Definition
Feature Extraction:	
$M(\cdot)$	The operation of mapping for unifying dimensions.
$\mathcal{M}(\cdot)$	The operation of masking non-central pixels of patch.
γ	The neighboring pixel decorrelation ratio.
$F(\cdot)$	The feature extraction operation of NRMFF.
$\mathcal{F}(\cdot)$	All operations for feature extraction.
Prototype Mask Contrast:	
$p_{t\setminus s}^k$	The representative prototype of the k-th class, and the subscript $t\setminus s$ refers to the TD\SD.
$f_\theta(\cdot)$	The feature extractor differs between the two training stages.
μ_t^j, σ_t^j, $\mathcal{T}_t^j$	The mean, standard deviation, and threshold interval of the j-th feature channel.
$\mathcal{K}_t\setminus\mathcal{R}_t$	The key\redundant mask group matrix.
$\mathcal{K}_t^{:,j}\setminus\mathcal{R}_t^{:,j}$	The key\redundant mask of the j-th feature channel.
$\mathcal{K}_t^{i,:}\setminus\mathcal{R}_t^{i,:}$	The key\redundant mask of p_t^i.
$\mathcal{C}(\cdot)$	The ratio correction operator.
β	The prototype mask correction factor.
g_t, G_t	The prototype group set and its matrix representation.
$p_t^{(i+)}$, $p_t^{(i-)}$	The positive and negative pairs generated by p_t^i.
$p_t^{i'}$, g_t'	The contrastive features of p_t^i after GC, and the set of contrastive features formed by all elements in g_t.
$\mathcal{N}_{t,i}^{must\setminus candidate}$	The must-consider\candidate negative pair set of $p_t^{i'}$.
$d(\cdot,\cdot)$	The Euclidean distance between two features.
Cross-Stage Feature Refinement:	
$\mathcal{MLP}(\cdot)$	The MLP refines features by focusing on key components across domains to form new-level features.
$\cos(\cdot,\cdot)$	The cosine similarity between two features.
η	The parameter adjusts the intensity of inter-class or intra-class distances.
λ	The prototype L2 norm regularization coefficient.
s	The interval between the two training stages.
P_t^k	The prototype of the k-th class for FSL classification.

A. PMC-CFR Framework

The flowchart of PMC-CFR is depicted in Fig. 1. The feature extraction module is built upon NRMFF, fusing features inclined toward different dimensional information after randomly masking the neighboring pixels of a patch. The PMC module employs PAMG to identify components exhibiting commonality among prototypes as potential redundancy to form negative pairs, while treating the remaining components as key features to form positive pairs. After further extracting contrastive features via GC, the dynamic triplet loss is optimized to achieve SCL. Furthermore, the cross-stage feature refinement module is coordinated by MLP and DC through iterative two-stage training. In the second stage, the shared MLP fits positive pairs to disregard potential redundancy in each domain, generating new features at a different level from the first stage. Across different levels in both stages, the DC separates interclass prototypes under directional guidance and reduces intraclass distances, while multilevel features are utilized to optimize the FSL classification loss. The key notations used in PMC-CFR are shown in Table I.

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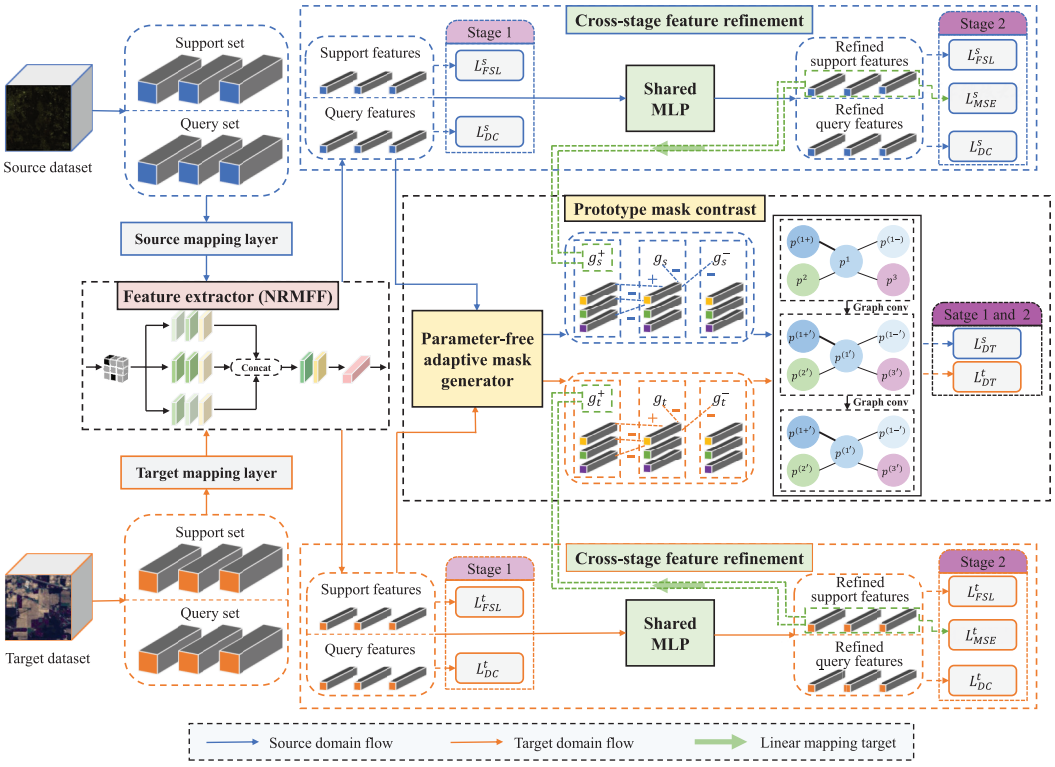


Fig. 1. Overall framework of PMC-CFR, which comprises three modules: feature extraction, PMC, and cross-stage feature refinement. Stage 1 represents the first training stage, while stage 2 represents the second training stage. The optimization losses differ between the two stages.

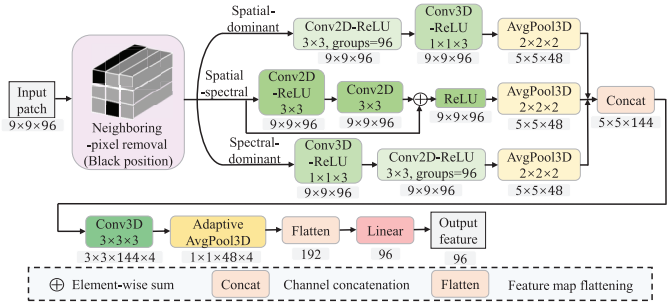


Fig. 2. Overall structure of NRMFF. The figure also illustrates the output size after each operation.

B. Feature Extraction Based on NRMFF

The training process alternates between the SD and TD, using two cross-domain nonshared mapping layers for cross-domain dimension alignment [45]. Each mapping layer is implemented by a 2-D convolutional layer with b filters of size 1×1. Denote the input HSI patch as $x \in \mathbb{R}^{9\times9\times c}$, where c represents the number of spectral bands and 9×9 denotes the spatial dimension. Let the mapping operation be denoted as $M(\cdot)$. The processed output is $x' = M(x) \in \mathbb{R}^{9\times9\times b}$, where b is the unified dimension. x' is then fed into NRMFF.

The structure of NRMFF is shown in Fig. 2. To avoid information loss and retain subtle differences for CL, b is set to 96, and the parameters of each layer are appropriately configured to ensure the output dimension m remains consistent with the input dimension b. Up to γ proportion of neighboring pixels in x' are randomly masked to reduce dependence on neighboring pixels during feature extraction. This mask operation is denoted as $\mathcal{M}(\cdot)$. Next, parallel branches are used to extract features with different tendencies in spatial and spectral dimensions. The spatial-dominant branch first uses grouped 2-D convolution with b groups and 3×3 filter size to extract local spatial information within each spectral channel, followed by a $1 \times 1 \times 3$ 3-D convolution kernel, which relies on spatial features to supplement local spectral information for each pixel. This convolutional sequence ensures that spatial information dominates feature representation. The convolutional structure of the spectral-dominant branch is the opposite of the spatial-dominant branch, where spectral information instead dominates feature representation. The spatial–spectral branch consecutively applies two 2-D convolutional layers, each with b filters of size 3×3, associating features with both spatial and spectral information while incorporating a residual structure. The outputs of the three branches are concatenated along the channel dimension after 3-D average pooling. They are then passed through a 3-D convolution layer with four $3 \times 3 \times 3$ kernels, followed by 3-D adaptive average pooling and fully connected layers to fully integrate features characterized by different dimensional inclinations. Let the feature extraction process in NRMFF be denoted as $F(\cdot)$. The feature $Y \in \mathbb{R}^m$ of x can be obtained as follows:

$$Y = F\left(\mathcal{M}\left(M\left(x\right), \gamma\right)\right). \tag{1}$$

For simplicity, all operations of the mapping layer and NRMFF are collectively denoted as $\mathcal{F}(\cdot)$.

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C. Prototype Mask Contrast

1) Construction of Positive and Negative Pairs Based on PAMG: The features of all samples in S_t and Q_t are considered to reduce sample fluctuations across different meta-tasks. The representative prototype p_t^k of the kth class in the TD is calculated as

$$p_t^k = \frac{1}{K+N} \sum_{x_t^j \in (S_t^k \cup Q_t^k)} f_\theta\left(x_t^j\right) \tag{2}$$

where S_t^k and Q_t^k are the sample sets of the kth class in S_t and Q_t, respectively. $f_\theta(\cdot)$ is the feature extractor with parameters θ, differing between the two training stages. However, the implementation of this module remains identical in both stages, with p_t^k consistently extracted using $f_\theta(\cdot) = \mathcal{F}(\cdot)$.

The C representative prototypes from all classes are collectively referred to as a prototype group, denoted as $g_t = \{p_t^1, p_t^2, \ldots, p_t^C\}$. PAMG evaluates the distribution of g_t over the m feature channels. The mean μ_t^j and standard deviation σ_t^j of the jth feature channel are calculated as

$$\mu_t^j = \frac{1}{C} \sum_{k=1}^{C} p_t^k\left[j\right] \tag{3}$$

$$\sigma_t^j = \sqrt{\frac{1}{C} \sum\nolimits_{k=1}^{C} \left(p_t^k\left[j\right] - \mu_t^j\right)^2} \tag{4}$$

where $p_t^k[j]$ is the jth feature component of p_t^k.

We identify potential redundant channels based on the statistical distribution of prototypes across the same feature channel. Specifically, we define the threshold interval $\mathcal{T}_t^j = (\mu_t^j - \sigma_t^j, \mu_t^j + \sigma_t^j)$ for the jth feature channel, which characterizes the average response range of this channel across different class prototypes. If the value of a certain class on this channel falls within this interval, it indicates proximity to the overall mean and relatively low variation across certain classes. From the perspective of class discriminability, such channels exhibit high consistency among certain classes and are more likely to carry redundant shared or weakly discriminative information, which can lead to classification confusion or redundancy dependence. Therefore, they are regarded as potential redundant channels, and their features are selected by redundant masks. In contrast, channels with values falling outside the corresponding interval tend to show higher interclass variation and discriminability. Thus, they are considered key channels, with their features selected by key masks. Although this threshold interval does not explicitly quantify interclass separability, it can serve as an approximate indicator for evaluating channel response consistency and discriminative potential. Each mask selects the specified features while masking the others. The key mask $\mathcal{K}_t^{:,j} \in \mathbb{R}^C$ and the redundant mask $\mathcal{R}_t^{:,j} \in \mathbb{R}^C$ of the jth feature channel are generated as follows:

$$\mathcal{K}_t^{:,j} = \begin{cases} \mathcal{K}_t^{:,j}[i] = 1, & p_t^i\left[j\right] \notin \mathcal{T}_t^j \\ \mathcal{K}_t^{:,j}[i] = 0, & p_t^i\left[j\right] \in \mathcal{T}_t^j \end{cases} \quad i = 1, 2, \ldots, C \tag{5}$$

$$\mathcal{R}_t^{:,j} = 1 - \mathcal{K}_t^{:,j} \tag{6}$$

where the superscript $:, j$ in $\mathcal{K}_t^{:,j}$ or $\mathcal{R}_t^{:,j}$ indicates that the channel mask is the jth column vector of the key mask group matrix $\mathcal{K}_t$ or the redundant mask group matrix $\mathcal{R}_t$.

The key masks and the redundant masks of all channels can be obtained in the same manner, thus generating $\mathcal{K}_t \in \mathbb{R}^{C\times m}$ and $\mathcal{R}_t \in \mathbb{R}^{C\times m}$, which contain the masks of all representative prototypes. The key mask and the redundant mask of p_t^i are denoted as $\mathcal{K}_t^{i,:} \in \mathbb{R}^m$ and $\mathcal{R}_t^{i,:} \in \mathbb{R}^m$, where the superscript $i,:$ in $\mathcal{K}_t^{i,:}$ or $\mathcal{R}_t^{i,:}$ indicates that the prototype mask is the ith row vector of $\mathcal{K}_t$ or $\mathcal{R}_t$. In $\mathcal{K}_t^{i,:}$, ones mean the retention of key components, while zeros indicate the masking of potential redundant components, thus forming a positive pair. $\mathcal{R}_t^{i,:}$ complementarily retains the potential redundant components and masks out the key components to form a negative pair.

Due to the unpredictable distribution of each feature channel, certain prototypes may be excessively masked by the key masks or overly retained by the redundant masks, leading to the generation of extreme contrastive samples. Therefore, to prevent excessive features from being identified as potential redundant components, ratio correction is applied to each prototype mask based on β, where β is defined as the prototype mask correction factor. The ratio of zeros in $\mathcal{K}_t^{i,:}$ and ones in $\mathcal{R}_t^{i,:}$ is the same, reflecting the ratio of potential redundancy. The ratio is computed and checked whether it is less than β. If the ratio exceeds β, zero in $\mathcal{K}_t^{i,:}$ is randomly selected to set to one, and one in $\mathcal{R}_t^{i,:}$ is randomly selected to set to zero, until the ratio falls below β. This correction mechanism does not rely on the specific response values of individual channels, thereby avoiding the model's tendency to develop dependency on correction adjustments for specific channels during training. Instead, it adjusts part of the mask bits in a balanced manner. Although the strategy of random selection introduces a certain degree of noise perturbation, on the one hand, the correction ratio is strictly constrained by β, limiting the extent of mask adjustment. The corrected masks still remain predominantly characterized by their original properties, thereby preventing the semantic degradation of masks and contributing to a dynamic balance in the feature allocation between positive and negative pairs. On the other hand, the model's adaptation to such uncertain perturbations helps enhance its generalization performance across different meta-tasks. The ratio correction for $\mathcal{K}_t^{i,:}$ and $\mathcal{R}_t^{i,:}$ is expressed as follows:

$$\mathcal{K}_t^{i,:}, \mathcal{R}_t^{i,:} = \mathcal{C}\left(\mathcal{K}_t^{i,:}, \mathcal{R}_t^{i,:}, \beta\right) \tag{7}$$

where $\mathcal{C}(\cdot)$ represents the ratio correction operator, including computing the mask ratio, comparing it with β, and performing random adjustments when the ratio exceeds β.

Notably, since $\mathcal{K}_t^{i,:}$ and $\mathcal{R}_t^{i,:}$ are complementary, uniformly applying β to correct them, rather than setting them separately, helps maintain the balance of feature selection while ensuring the contrast between positive and negative pairs. The construction of positive and negative pairs for p_t^i is illustrated in Fig. 3. Finally, each element of g_t is sequentially treated as a row vector to form its matrix representation $G_t \in \mathbb{R}^{C\times m}$. The positive contrastive group matrix G_t^+ and the negative contrastive group matrix G_t^- are computed as follows:

$$G_t^+ = G_t \odot \mathcal{K}_t \tag{8}$$

$$G_t^- = G_t \odot \mathcal{R}_t \tag{9}$$

where $\odot$ denotes the Hadamard product.

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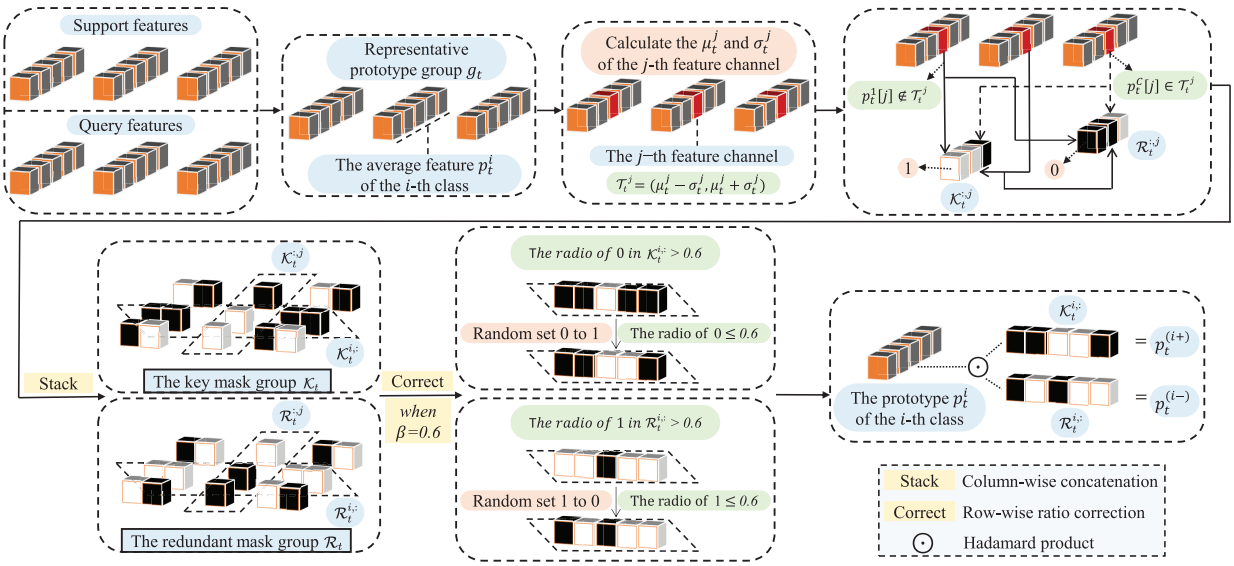


Fig. 3. Overall process of constructing the positive and negative pairs for p_t^i using PAMG. The color of the blocks in the masks represents the corresponding values, where white or black indicates a bit value of 1 or 0. It is necessary to compute the masks for all channels and perform ratio correction on all prototype masks. Only a single example is shown in the figure for each case, and β is set to 0.6 to illustrate the ratio correction process for the ith prototype mask.

Each row of G_t^+ and G_t^- is sequentially treated as an element, resulting in $g_t^+ = \{p_t^{(1+)}, p_t^{(2+)}, \ldots, p_t^{(C+)}\}$ and $g_t^- = \{p_t^{(1-)}, p_t^{(2-)}, \ldots, p_t^{(C-)}\}$.

2) Dynamic Triplet Loss: The prototype contrast group is denoted as $g_{t,c} = g_t \cup g_t^+ \cup g_t^-$, and its matrix form is represented as $G_{t,c} \in \mathbb{R}^{3C \times m}$. The ith element in $g_{t,c}$ is denoted as the node v_t^i, and the element A_t^{ij} in the ith row and jth column of the adjacency matrix $A_t \in \mathbb{R}^{3C \times 3C}$ is calculated using the radial basis function (RBF) as follows:

$$A_t^{ij} = \exp\left(-\frac{\left\|v_t^i - v_t^j\right\|_2^2}{2\mathcal{Q}^2}\right) \tag{10}$$

where $\mathcal{Q}$ is the parameter controlling the width of the RBF, and $\|\cdot\|_2$ denotes the two-norm of a vector.

To preserve the information of nodes during the GC process, a self-loop is typically added to A_t, resulting in $\tilde{A}_t = A_t + I$, where I is the identity matrix of the corresponding size. $\tilde{D}_t$ is the degree matrix of $\tilde{A}_t$, defined as $\tilde{D}_t^{ii} = \sum_j \tilde{A}_t^{ij}$, where $\tilde{D}_t^{ii}$ represents the degree of v_t^i. GC performs layer propagation using the normalized adjacency matrix as follows:

$$H_t^{(l)} = \sigma\left(\tilde{D}_t^{-\frac{1}{2}} \tilde{A}_t \tilde{D}_t^{-\frac{1}{2}} H_t^{(l-1)} W^{(l)}\right) \tag{11}$$

where $H_t^{(l)}$ and $W^{(l)}$ are the output and the shared weight matrix of the lth layer, respectively, $H_t^{(0)} = G_{t,c}$ is the input of the first layer, and σ is the ReLU activation function.

The output of the lth GC layer is denoted as $G'_{t,c} = H_t^{(l)}$. Each row of $G'_{t,c}$ serves as the new representation for the corresponding feature, resulting in the set $g'_{t,c} = g'_t \cup g_t^{(+')} \cup g_t^{(-')}$. Each element $p_t^{(i')}$ in g'_t is used as the anchor of the ith class. For $p_t^{(i')}$, $p_t^{(i+')}$ in $g_t^{(+')}$ is its only positive pair, while all other elements in $g'_{t,c}$ serve as its negative pairs. $p_t^{(i-')}$ in $g_t^{(-')}$ reflects the potential redundant features of the ith class, whereas the anchors of other classes in g'_t are crucial for classification. These representative negative pairs must be considered, denoted as the set $\mathcal{N}_{t,i}^{\text{must}}$. The remaining negative pairs are generated based on anchors from other classes. As derived features, although they share certain similarities with the original features to some extent, they may also serve a complementary role. Therefore, they are considered as candidate negative pairs, denoted as the set $\mathcal{N}_{t,i}^{\text{candidate}}$. Only the n closest pairs in $\mathcal{N}_{t,i}^{\text{candidate}}$ are selected based on distance for CL, which also helps prevent the critical impact of close derived negative pairs from being diluted by distant ones. The dynamic triplet loss, as defined in (12), pulls positive pairs together and separates negative pairs, thereby improving the model's capability to shape discriminative features

$$L_{\text{DT}}^t = \frac{1}{C}\sum_{i=1}^{C} \max\left(d\left(p_t^{(i')}, p_t^{(i+')}\right) - d_{\text{neg}}\left(p_t^{(i')}, p_t^{(j')}\right) + \alpha_i, 0\right) \tag{12}$$

$$d_{\text{neg}}\left(p_t^{(i')}, p_t^{(j')}\right) = \frac{1}{C+n}\left(\sum_{p_t^{(j')} \in \mathcal{N}_{t,i}^{\text{must}}} d\left(p_t^{(i')}, p_t^{(j')}\right) + \sum_{p_t^{(j')} \in \mathcal{N}_{t,i}^{\text{candidate}}}^{\text{minest:n}} d\left(p_t^{(i')}, p_t^{(j')}\right)\right) \tag{13}$$

$$\alpha_i = \rho_{\text{base}} + \rho_{\text{scale}} \times \left(d_{\text{neg}}\left(p_t^{(i')}, p_t^{(j')}\right) - d\left(p_t^{(i')}, p_t^{(i+')}\right)\right) \tag{14}$$

where $\max(\cdot,\cdot)$ denotes the maximum value of the inputs, and $d(\cdot,\cdot)$ represents the Euclidean distance. The negative pair distance of $p_t^{(i')}$ is defined in (13), where $\sum_{p_t^{(j')} \in \mathcal{N}_{t,i}^{\text{candidate}}}^{\text{minest:n}} d(p_t^{(i')}, p_t^{(j')})$ denotes the sum of the distances

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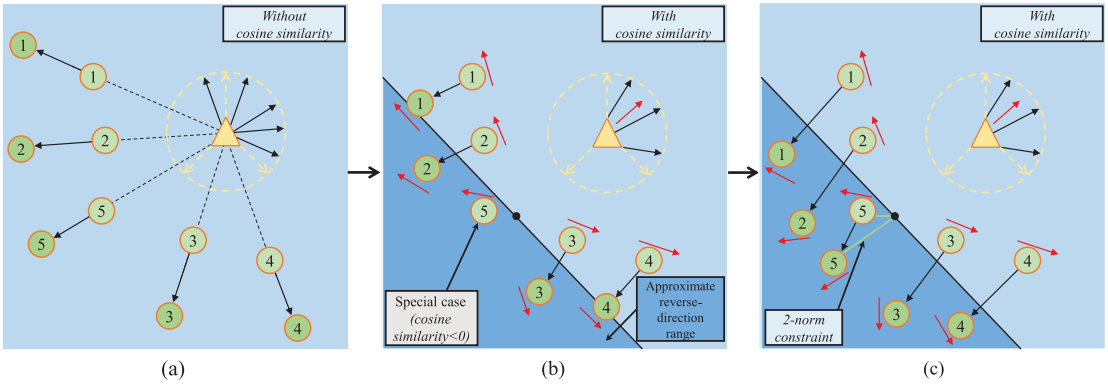


Fig. 4. Stepwise improvement effects after applying the interclass DC. The yellow prototype indicates that the distribution of this class is stable across different meta-tasks, and the region inside the yellow circle is used to approximate the variation range of this class prototype across different meta-tasks and during interclass calibration. The green prototypes represent that the distribution of this class exhibits significant fluctuation across different meta-tasks, distinguished by numbers to indicate their origin from different meta-tasks. The red arrow indicates the direction of the prototype. (a) Issues arising from prototype calibration based solely on distance. (b) Special case of prototype calibration with directional guidance after introducing cosine similarity. (c) Resolving the uncalibrated special case by introducing the two-norm constraint on prototypes.

for the n closest negative pairs in $\mathcal{N}_{t,i}^{\text{candidate}}$. In (14), α_i is the margin that controls the minimum distance difference between the positive and negative pairs of $p_t^{(i')}$. ρ_{base} and ρ_{scale} adaptively adjust α_i according to the distance difference between the positive and negative pairs.

D. Cross-Stage Feature Refinement

1) MLP-Based Feature Refinement Driven by Positive Pairs: The positive pair of each representative prototype masks potential redundant components, aiming to yield a more refined discriminative property. The shared MLP is introduced to refine cross-domain features by focusing on key components in each domain through fitting positive pairs, denoted as $\mathcal{MLP}(\cdot)$. The features in S_t are mapped through the MLP to fit the positive pairs of the corresponding class for mean squared error (mse) optimization as follows:

$$L_{\text{mse}}^t = \frac{1}{C \times K} \sum_{k=1}^{C} \sum_{x_t^i \in S_t^k} \frac{1}{m} \sum_{j=1}^{m} \left(\mathcal{MLP}\left(\mathcal{F}\left(x_t^i\right)\right)[j] - p_t^{(k+)}[j] \right)^2. \tag{15}$$

Based on this, an iterative two-stage training mode is established, where the feature refinement via MLP is applied only in the second stage to extract a new level of features. The MLP further exploits the intrinsic properties of contrastive features to strengthen feature representativeness by perceiving key features within each domain. The structure of the MLP is lightweight, consisting of only two fully connected layers. The feature channel size is first reduced by half and then restored, with GELU activation function applied between the two layers.

2) Directional Calibration: The distribution fluctuations of the same class across different meta-tasks may interfere with the effectiveness of interclass prototype separation. As shown in Fig. 4(a), when the prototype of one class exhibits stable distribution across different T_t (the yellow prototype), while the prototype of another class fluctuates significantly across different T_t (the green prototype), directly using Euclidean distance to separate the two class prototypes will cause the fluctuating prototypes to be pushed apart in multiple directions. For such a class with significant fluctuations, the calibration effects across multiple meta-tasks not only fail to accumulate but may also lead to certain antagonism. The Euclidean distance metric focuses on the sum of distance contributions from different feature components, which meets the expectation, but it ignores the directionality. The interclass DC loss is defined in (16), where cosine similarity is introduced into the Euclidean distance term to guide the direction. As shown in Fig. 4(b), by considering the mutual balance between distance and direction, the fluctuating prototypes tend to converge to a closer range near the opposite interclass direction. However, incorporating cosine similarity weakens the influence of the distance metric, and for prototypes already within the reverse-direction range, the calibration loss may even disregard the effect of the distance metric. To address this problem, the two-norm constraint on prototypes is added in (16). As shown in Fig. 4(c), for prototypes already within the reverse-direction range, the constraint term ensures sufficiently large prototype norms, indirectly guaranteeing adequate interclass distance. In other cases, the interference of prototype norm constraint on the distance metric instead facilitates the balanced optimization of direction and distance. Note that since the class order in meta-task construction is randomized, the two-norm imposes random constraint strengths for each class across different meta-tasks, which enhances adaptability to task variations. The interclass DC aims to learn generalizable interclass separable features across different meta-tasks, encouraging features from the same class to be distributed within a range as close as possible, guided by direction

$$L_{\text{inter}}^t = \frac{2}{C(C-1)} \sum_{i=1}^{C} \sum_{j=i+1}^{C} \left(\frac{\cos\left(p_t^i, p_t^j\right)}{\exp\left(\eta d\left(p_t^i, p_t^j\right)\right)} + \lambda \exp\left(-\|p_t^i\|_2\right) \right) \tag{16}$$

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where $\cos(\cdot,\cdot)$ denotes the cosine similarity, η is used to adjust the intensity of interclass or intraclass distances, and λ controls the influence of the two-norm constraint term.

With the directional range established for same-class feature distribution, the intraclass distances are minimized using (17). Only the n farthest samples are chosen for each class to calculate the loss. Setting n to a small value can focus on optimizing the intraclass compactness from extreme samples, thereby avoiding the risk of overfitting caused by excessive smoothing of the global distribution

$$L_{\text{intra}}^{t} = \frac{1}{C}\sum_{k=1}^{C}\sum_{x_t^j \in \left(S_t^k \cup Q_t^k\right)}^{\text{maxest:n}} \log\left(\eta^{-1} d\left(f_\theta\left(x_t^j\right), p_t^k\right) + 1\right) \tag{17}$$

$$L_{\text{DC}}^{t} = L_{\text{inter}}^{t} + L_{\text{intra}}^{t} \tag{18}$$

where $\sum_{x_t^j \in (S_t^k \cup Q_t^k)}^{\text{maxest:n}} \log(\eta^{-1} d(f_\theta(x_t^j), p_t^k) + 1)$ represents the sum of the n farthest intraclass distances for the kth class. The overall DC loss is defined in (18).

3) Two-Stage Training Mode: The proposed method adopts an iterative two-stage training mode. Let the interval between the two stages be denoted as s. For every $2s$ training episodes, the first s episodes are designated as the first stage and the subsequent s episodes as the second stage. The second stage employs the shared MLP, driven by positive pairs, to focus on key components in each domain for feature refinement. The effect of CL in the first stage supports the refinement of the MLP, while the CL applied again in the second stage further assists the MLP in enhancing feature discriminability, resulting in a new feature level distinct from the first stage. The DC further refines features across different levels in two stages under directional guidance. The purpose of the two-stage training is to progressively refine feature representations hierarchically. Through iterative two-stage training, the MLP performs hierarchical feature modeling, while the DC facilitates semantic separation. Their synergy enables cross-stage feature refinement, allowing the model to continuously revisit and refine features at different levels to fully exploit their potential. Overall, this training mode provides advantages unattainable in single-stage training, and better integrates the complementary effects of all modules, enabling more fine-grained feature modeling and transfer learning, ultimately promoting cross-domain feature distribution alignment.

Each training stage includes the FSL process. The prototype P_t^k of the kth class in FSL is calculated as

$$P_t^k = \frac{1}{K}\sum_{x_t^j \in S_t^k} f_\theta\left(x_t^j\right). \tag{19}$$

The class distribution $\mathcal{P}_\theta^t(\cdot)$ for each sample x_t^j in Q_t can be expressed as

$$\mathcal{P}_\theta^t\left(y_t^j = k \middle| x_t^j \in Q_t\right) = \frac{\exp\left(-d\left(f_\theta\left(x_t^j\right), P_t^k\right)\right)}{\sum_{k=1}^{C}\exp\left(-d\left(f_\theta\left(x_t^j\right), P_t^k\right)\right)}. \tag{20}$$

The FSL loss of each training stage is as follows:

$$L_{\text{FSL}}^{t} = E_{S_t, Q_t}\sum_{\left(x_t^j, y_t^j\right)\in Q_t} -\log \mathcal{P}_\theta^t\left(y_t^j = k \middle| x_t^j\right). \tag{21}$$

The two training stages use different $f_\theta(\cdot)$ to calculate the representative prototypes for the DC loss and the prototypes for the FSL loss. In the first stage, $f_\theta(\cdot) = \mathcal{F}(\cdot)$, while in the second stage, $f_\theta(\cdot) = \mathcal{MLP}(\mathcal{F}(\cdot))$. Both stages optimize the dynamic triplet loss, but only the second stage optimizes the mse. The total losses L_1^t and L_2^t to be optimized in the first and second stages are, respectively, defined as follows:

$$L_1^t = L_{\text{DT}}^t + L_{\text{FSL}}^t + L_{\text{DC}}^t \tag{22}$$

$$L_2^t = L_{\text{DT}}^t + L_{\text{FSL}}^t + L_{\text{DC}}^t + L_{\text{mse}}^t. \tag{23}$$

Algorithm 1 Training Process of PMC-CFR

Input: S_t and Q_t in $\mathbf{D}_t$, S_s and Q_s in $\mathbf{D}_s$, $\mathcal{F}(\cdot)$ and $\mathcal{MLP}(\cdot)$ related to feature extraction, the number of training episodes, and the staged training interval s.

Output: The loss of the model.

```
for episode = 1: episodes do
    if episode % 2 = 0 then
        Extract features Y from S_t and Q_t using F(·);
        Perform the prototype mask contrast process based on Y to calculate L_DT^t by Eq.(12);
        if episode % 2s < s then
            Calculate L_DC^t through Y by Eq.(18);
            Calculate L_FSL^t through Y by Eq.(21);
            Return L_1^t of an episode by Eq.(22).
        else
            Extract new features Y' from Y using MLP(·) and calculate L_mse^t by Eq.(15);
            Calculate L_DC^t through Y' by Eq.(18);
            Calculate L_FSL^t through Y' by Eq.(21);
            Return L_2^t of an episode by Eq.(23).
        end if
    else
        Train on S_s and Q_s in the same way as above.
    end if
end for
```

Meta-task training is conducted alternately between the SD and TD. Similarly, the two training stages optimize the total losses L_1^s and L_2^s in the SD, respectively. The detail of the training process is shown in Algorithm 1. The classification process during the testing phase is illustrated in Fig. 5. TD features are extracted using $f_\theta(\cdot) = \mathcal{MLP}(\mathcal{F}(\cdot))$, and a KNN classifier is trained on the features of a few labeled samples to predict the labels of the unlabeled samples.

III. Experiments and Analysis

A. Experimental Datasets

The experiments select the Chikusei dataset, which contains rich class information, as the SD dataset. The Indian Pines (IP), University of Pavia (UP), Houston, and Salinas (SA) datasets are chosen as the TD datasets. Figs. 6–10 show the pseudo-color images and ground-truth maps of the five datasets, where the numbers following the class names in the color bars of the TD datasets indicate the correspondence between class names and class indices.

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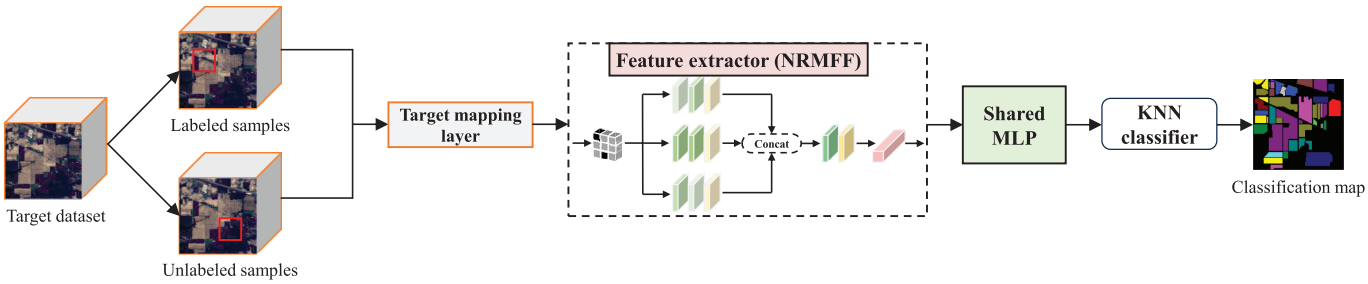


Fig. 5. Flowchart of the testing phase for the proposed PMC-CFR.

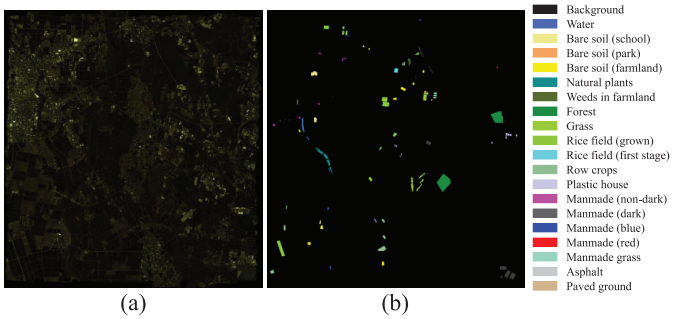


Fig. 6. Chikusei dataset. (a) Pseudo-color image. (b) Ground-truth map.

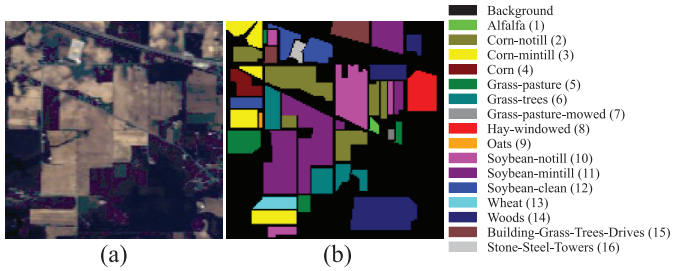


Fig. 7. IP dataset. (a) Pseudo-color image. (b) Ground-truth map.

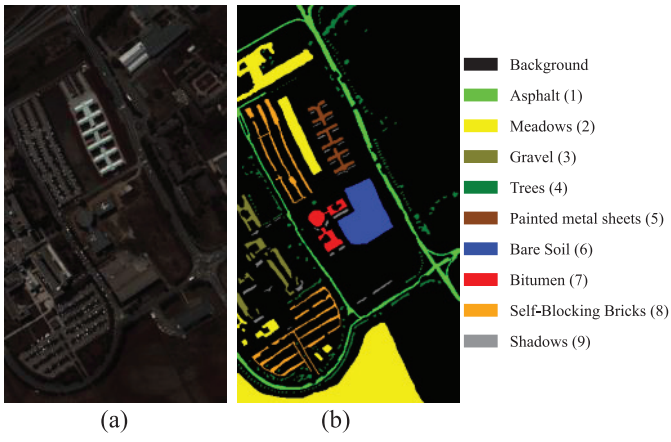


Fig. 8. UP dataset. (a) Pseudo-color image. (b) Ground-truth map.

1) SD Dataset: The Chikusei dataset was collected by the hyperspectral visible/near-infrared cameras (Hyperspec-VNIR-C) sensor in Chikusei, Japan. This dataset contains 128 bands with a wavelength range from 363 to 1018 nm. It has a spatial resolution of 2.5 m with 2517 × 2335 pixels and 19 classes.

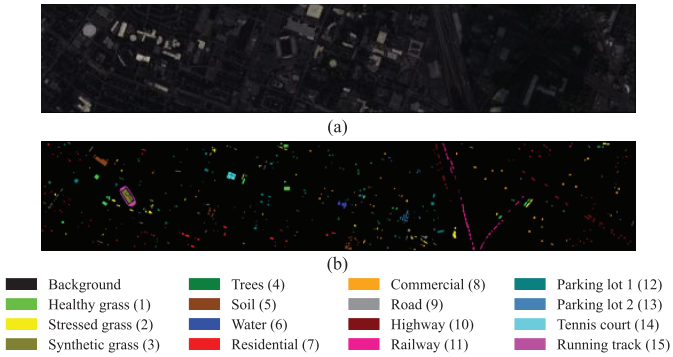


Fig. 9. Houston dataset. (a) Pseudo-color image. (b) Ground-truth map.

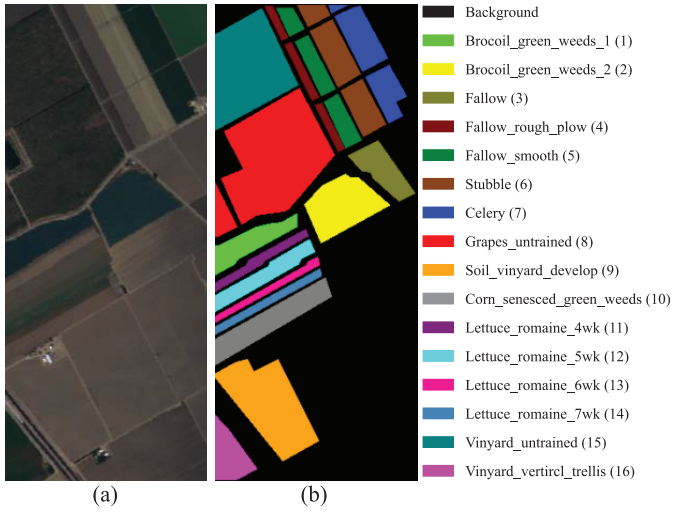


Fig. 10. SA dataset. (a) Pseudo-color image. (b) Ground-truth map.

2) TD Dataset: The IP dataset was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor at the Indiana Pines region of the United States, which has a spatial resolution of 20 m with 145 × 145 pixels and 16 classes. It contains 200 bands with a wavelength range from 400 to 2500 nm after removing 20 noise-affected bands.

The UP dataset was acquired by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor in Pavia, Italy, which has a spatial resolution of 1.3 m with 610 × 340 pixels and nine classes. It includes 103 bands with a wavelength

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range from 430 to 860 nm after removing 12 noise-affected bands.

The Houston dataset, captured by the Compact Airborne Spectrographic Imager (CASI) sensor in a suburban area of Houston, TX, USA, contains 144 bands with a wavelength range from 380 to 1050 nm. It has a spatial resolution of 2.5 m with 349 × 1905 pixels and 15 classes.

The SA dataset was collected by the AVIRIS sensor in Salinas Valley, CA, USA, containing 204 bands after removing 20 invalid bands. This dataset spans a wavelength range from 400 to 2500 nm. It has a spatial resolution of 3.7 m with 512 × 217 pixels and 16 classes.

B. Experimental Setting

All experiments were conducted using the PyTorch framework on a platform with an i9-12900HX CPU, NVIDIA GeForce RTX 4060 16GB GPU, and 16GB of memory. As a cross-domain FSL method, 200 labeled samples per class are selected from the Chikusei dataset for training, while only five labeled samples per class are chosen from the target dataset for training. Each C-way K-shot T is treated as one iteration, where C is set to the number of classes in the TD dataset, and K is set to 1. The number of samples per class N in Q is set to 19. The spatial size of the input patch is 9×9, the number of GC layers l is set to 3, Q is set to 0.1 for calculating the adjacency matrix, and the interval s between the two training stages is set to 500. The number of nearest neighbors used in the KNN classifier for classification is set to 1. The number of samples n selected for loss computation in both the dynamic triplet loss and DC loss is uniformly set to 3. ρ_{base} and ρ_{scale} for dynamically adjusting α_i in the dynamic triplet loss are set to 8 and 0.5, respectively, while η for adjusting the intensity of interclass or intraclass distances in the DC loss is set to 0.5. The number of training episodes is set to 10 000, and the Adam optimizer is used. The learning rate is set to 0.0005 and decays by 20% every 4 000 iterations.

We use overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) to evaluate the classification performance of different networks. Each experiment is independently run ten times, and the average result along with the standard deviation is reported.

C. Parameter Tuning

The experiments were conducted to fine-tune the neighboring pixel decorrelation ratio γ, the prototype mask correction factor β, and the prototype L2 norm regularization coefficient λ to assess their impact on PMC-CFR. For γ and β, which directly influence feature extraction, we perform a combination experiment where γ is selected from {0, 0.05, 0.15, 0.3} and β is selected from {0.4, 0.5, 0.6, 0.7}. For λ, which affects network optimization, it is selected from {0, 0.05, 0.1, 0.2, 0.4}. The values of γ and λ are set at unequal intervals to explore the impact of different degrees. To alleviate the pressure of parameter tuning, based on the experimentally obtained suboptimal parameters, we first set γ as {0.15, 0.05, 0.15, 0.3} and β as {0.6, 0.6, 0.6, 0.6} for the IP, UP, Houston, and SA datasets, respectively, and conduct experiments targeting λ.

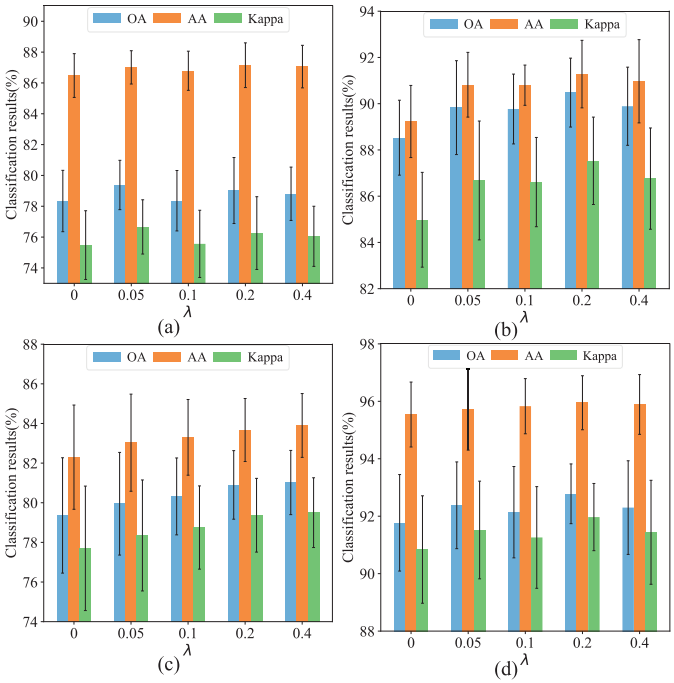


Fig. 11. Quantitative evaluation of the prototype L2 norm regularization coefficient λ on four datasets. (a) IP. (b) UP. (c) Houston. (d) SA.

The classification results under different values of λ are shown in Fig. 11. A larger λ enhances the correction of the reverse-direction case during interclass DC, while introducing greater interference to the distance metric in other cases, increasing the demand for a more balanced optimization between direction and distance. When λ is set to 0, the classification performance of all datasets decreases due to the lack of the constraint. Based on the experimental results, the values of λ for the IP, UP, Houston, and SA datasets are set to 0.05, 0.2, 0.4, and 0.2, respectively. This indicates that the optimal adaptation strength of λ for different datasets primarily depends on the synergy between the internal feature distribution and the constraint term.

After setting the value of λ, Fig. 12 shows the variation trend of OA under different combinations of γ and β. It can be observed that appropriate neighboring-pixel removal can adapt to variations in adjacent terrain, stimulating better classification potential. The IP and Houston datasets tend to identify fewer potential redundant components for positive pairs, while the UP and SA datasets show the opposite trend. For the IP and Houston datasets, the decline of classification performance observed when γ is set to 0.3 and β to 0.7 deserves attention. This phenomenon indicates that when the decorrelation degree of neighboring pixels is high and the number of key components retained for positive pairs is too small, it may lead to a degradation in the quality of features. According to the experimental results, the values of γ for the IP, UP, Houston, and SA datasets are finally set to 0.3, 0.05, 0.15, and 0.15, and β to 0.4, 0.6, 0.4, and 0.6, respectively.

D. Comparison of Different MLP Architectures

To investigate the impact of MLP architecture complexity on the performance of PMC-CFR, we compare three MLP

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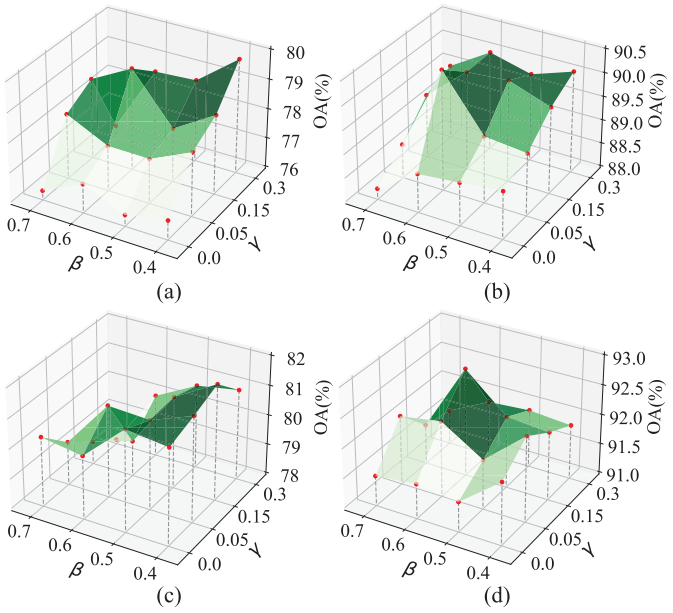


Fig. 12. Quantitative evaluation of different combinations of the neighboring pixel decorrelation ratio γ and the prototype mask correction factor β on four datasets. (a) IP. (b) UP. (c) Houston. (d) SA.

TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT MLP ARCHITECTURES

MLP Architecture	IP	UP	Houston	SA
Two layers (96 → 48 → 96)	**79.79**	**90.48**	**81.82**	**92.78**
Four layers (96 → 64 → 32 → 64 → 96)	77.81	88.23	80.39	91.65
Eight layers (96 → 64 → 32 → 64 → 96 → 128 → 160 → 128 → 96)	74.75	86.76	78.73	89.93

architectures with different depths, each with GELU activation between layers, as shown in Table II, which presents feature dimension changes across layers. It can be observed that as the number of MLP layers increases, classification performance gradually decreases across all datasets. This suggests that introducing an excessively deep MLP to fit cross-domain positive pairs with potential redundancy removed is detrimental to the learning of few-shot tasks in the TD. Overly complex mapping may cause the model to focus more on positive pairs generated from the abundant samples in the SD, leading to overfitting and ultimately impairing the classification features of the TD. In contrast, a lightweight MLP is sufficient to effectively refine cross-domain features for learning domain-invariant mapping.

E. Comparison With Other Methods

To evaluate the proposed PMC-CFR, it is compared with nine related models, including the SVM with RBF kernel [10], 3DCNN [12], DCFSL [23], Gia-CFSL [40], CMFSL [24], ADAFSL [26], DMCM [27], HyMuT [29], and GCC-FSL [41]. Among these, the SVM is a traditional machine learning method, 3DCNN is a traditional deep learning classification model, while the other models are cross-domain classification methods designed for few-shot scenarios in HSIs. All methods select five labeled samples per class from the target dataset for training in the same random manner. For the cross-domain

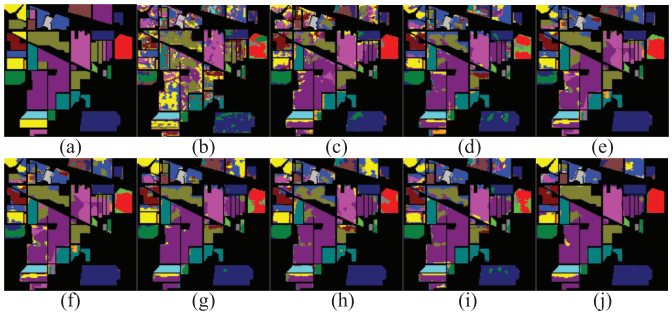


Fig. 13. Classification maps of each method on the IP dataset. (a) Ground truth. (b) 3DCNN (51.82%). (c) DCFSL (66.58%). (d) Gia-CFSL (66.40%). (e) CMFSL (67.49%). (f) ADAFSL (72.46%). (g) DMCM (74.41%). (h) HyMuT (70.53%). (i) GCC-FSL (72.01%). (j) PMC-CFR (79.79%).

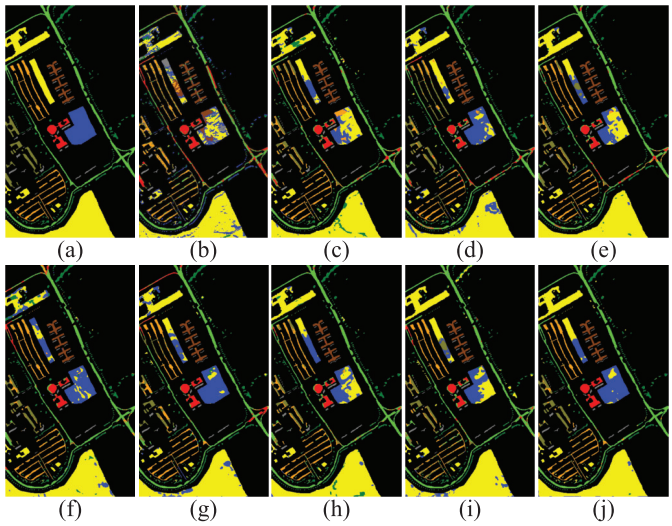


Fig. 14. Classification maps of each method on the UP dataset. (a) Ground truth. (b) 3DCNN (64.88%). (c) DCFSL (82.81%). (d) Gia-CFSL (82.99%). (e) CMFSL (83.79%). (f) ADAFSL (85.04%). (g) DMCM (86.43%). (h) HyMuT (84.96%). (i) GCC-FSL (85.83%). (j) PMC-CFR (90.48%).

classification methods, the Chikusei dataset is used as the SD dataset for transferring knowledge. The parameters for each method are set as per the original literature.

1) Performance on the IP Dataset: Table III lists the classification results on the IP dataset. The SVM and 3DCNN show relatively poor performance, while the meta-learning paradigm effectively improves classification performance. The proposed method demonstrates significant advantages over the other methods, achieving improvements of 5.38%, 4.67%, and 6.07% in OA, AA, and κ ($\times$100), respectively, compared to the second-best method. The classification accuracy achieves the best results over 14 classes, with improvements ranging from 8.14% to 13.95% in the challenging classes, including Corn-notill (class 2), Hay-windowed (class 8), and Soybean-clean (class 12). Fig. 13 shows the classification maps of different methods. The classification map of PMC-CFR is the smoothest. The high intraclass cohesion of the refined features ensures better integrity of the classified regions.

2) Performance on the UP Dataset: The classification results for the UP dataset are shown in Table IV. PMC-CFR still demonstrates the best performance, surpassing

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TABLE III

CLASSIFICATION RESULTS (%) OF EACH METHOD ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	DCFSL	Gia-CFSL	CMFSL	ADAFSL	DMCM	HyMuT	GCC-FSL	PMC-CFR
1	67.56	78.54	91.46	87.07	94.88	97.80	92.44	95.12	89.76	**99.27**
2	27.58	40.30	46.27	46.53	43.22	57.77	58.98	52.95	54.42	**69.21**
3	34.95	39.76	51.98	46.73	51.14	60.93	65.76	54.82	57.44	**66.48**
4	36.85	47.46	76.59	76.16	77.28	85.30	86.81	83.97	91.25	**94.27**
5	60.50	58.28	71.90	73.12	73.43	77.36	80.08	75.33	80.56	**83.83**
6	64.12	56.51	85.70	85.26	87.10	90.54	94.63	83.79	90.51	**97.38**
7	79.57	86.52	98.26	99.13	99.57	99.57	**100.00**	98.26	**100.00**	**100.00**
8	38.60	66.89	79.98	80.68	78.54	80.17	87.65	85.39	83.45	**95.79**
9	77.33	91.33	**100.00**	99.33	99.33	**100.00**	99.33	**100.00**	**100.00**	**100.00**
10	41.37	57.65	62.18	64.53	64.07	68.02	66.48	65.14	63.52	**72.90**
11	40.34	51.88	67.61	68.48	68.96	69.50	71.61	72.78	69.38	**75.45**
12	34.30	32.74	39.73	43.15	47.99	56.96	53.88	46.75	59.40	**73.35**
13	88.25	92.55	97.85	95.85	98.25	**99.40**	98.75	98.05	99.35	98.85
14	57.44	60.10	84.54	82.10	84.81	86.76	87.31	86.49	87.60	**88.90**
15	31.94	37.80	65.67	63.10	72.13	76.14	85.91	69.42	78.64	**87.87**
16	73.41	91.14	96.82	95.11	96.70	96.59	96.93	96.36	**98.30**	97.73
OA	43.63 ± 2.55	51.82 ± 4.16	66.58 ± 1.58	66.40 ± 1.95	67.49 ± 2.68	72.46 ± 2.67	74.41 ± 1.64	70.53 ± 1.88	72.01 ± 1.50	$\mathbf{79.79 \pm 1.65}$
AA	53.38 ± 1.71	61.84 ± 2.54	76.03 ± 1.26	75.39 ± 2.41	77.34 ± 1.98	81.43 ± 1.88	82.91 ± 1.71	79.04 ± 1.99	81.47 ± 1.89	$\mathbf{87.58 \pm 1.33}$
$\kappa \times 100$	37.29 ± 2.35	46.20 ± 4.26	62.22 ± 1.71	61.99 ± 2.09	63.27 ± 2.93	68.90 ± 2.97	71.06 ± 1.77	66.55 ± 2.13	68.37 ± 1.71	$\mathbf{77.13 \pm 1.84}$

TABLE IV

CLASSIFICATION RESULTS (%) OF EACH METHOD ON THE UP DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	DCFSL	Gia-CFSL	CMFSL	ADAFSL	DMCM	HyMuT	GCC-FSL	PMC-CFR
1	65.43	49.07	79.42	80.07	76.79	88.82	84.40	85.98	82.31	**92.28**
2	56.54	68.52	88.13	89.30	88.38	86.38	88.93	88.02	90.21	**90.83**
3	54.19	66.49	65.42	64.09	69.09	72.45	74.29	66.76	68.36	**77.53**
4	77.29	79.24	92.55	89.70	92.96	87.16	89.46	90.90	92.66	**94.66**
5	98.35	83.84	99.38	99.09	99.51	98.55	99.22	99.28	99.40	**99.90**
6	59.88	41.38	75.01	70.61	73.10	71.53	77.43	75.22	77.68	**84.01**
7	77.11	85.07	81.65	80.48	79.86	86.78	93.65	81.07	82.39	**96.48**
8	68.23	72.60	64.61	69.65	80.67	86.15	83.81	79.53	78.44	**91.76**
9	**99.83**	93.51	99.03	97.73	97.29	98.91	98.06	96.33	98.33	94.07
OA	63.59 ± 6.41	64.88 ± 6.24	82.81 ± 3.16	82.99 ± 2.07	83.79 ± 1.91	85.04 ± 2.57	86.43 ± 2.89	84.96 ± 2.41	85.83 ± 1.93	$\mathbf{90.48 \pm 1.49}$
AA	72.98 ± 2.67	71.08 ± 4.63	82.80 ± 2.20	82.30 ± 1.63	84.18 ± 1.88	86.30 ± 2.03	87.69 ± 2.68	84.79 ± 2.15	85.53 ± 1.52	$\mathbf{91.28 \pm 1.46}$
$\kappa \times 100$	54.81 ± 6.68	55.85 ± 6.90	77.56 ± 4.00	77.68 ± 2.64	78.77 ± 2.54	80.40 ± 3.28	82.19 ± 3.85	80.31 ± 3.00	81.44 ± 2.37	$\mathbf{87.53 \pm 1.89}$

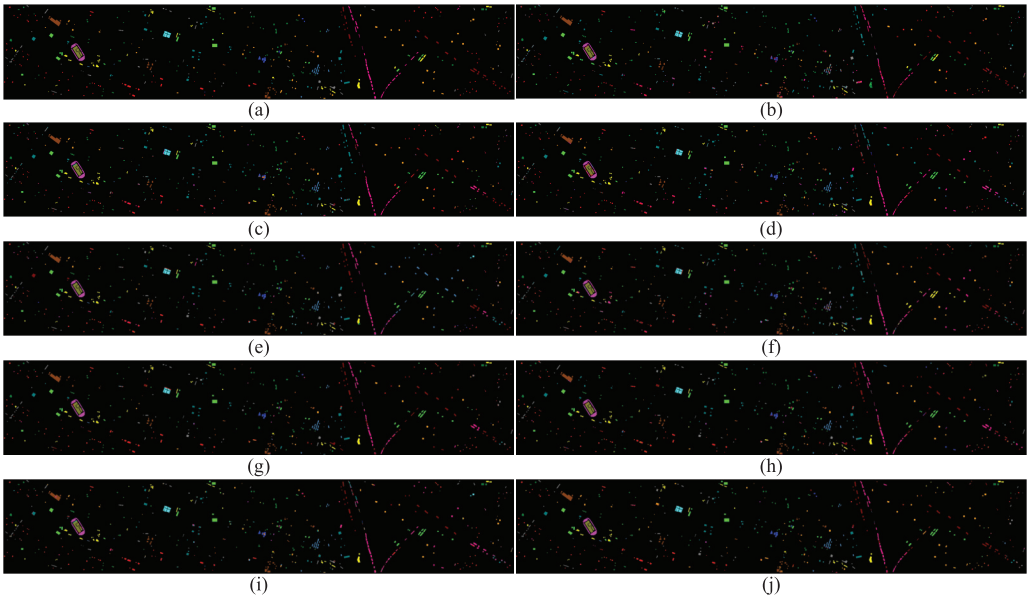


Fig. 15. Classification maps of each method on the Houston dataset. (a) Ground truth. (b) 3DCNN (68.82%). (c) DCFSL (76.17%). (d) Gia-CFSL (74.98%). (e) CMFSL (78.73%). (f) ADAFSL (74.92%). (g) DMCM (80.76%). (h) HyMuT (78.80%). (i) GCC-FSL (78.21%). (j) PMC-CFR (81.82%).

the second-best method by 4.05%, 3.59%, and 5.34% in OA, AA, and κ ($\times 100$), respectively. The proposed method achieves the highest classification accuracy in eight classes, with accuracy improvements of 6.33% and 5.61% on the

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TABLE V
CLASSIFICATION RESULTS (%) OF EACH METHOD ON THE HOUSTON DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	DCFSL	Gia-CFSL	CMFSL	ADAFSL	DMCM	HyMuT	GCC-FSL	PMC-CFR
1	86.67	91.14	93.13	92.34	93.06	93.42	**93.51**	93.39	92.64	90.61
2	67.77	80.43	84.44	81.33	84.60	85.30	**89.60**	84.82	85.93	89.41
3	95.90	93.79	95.91	95.25	94.08	97.50	97.40	97.23	96.16	**99.45**
4	78.86	86.61	91.68	91.50	92.80	92.69	93.17	92.70	91.21	**93.45**
5	73.89	92.09	94.46	92.38	94.68	95.97	96.90	97.52	94.99	**99.13**
6	82.63	65.31	83.22	78.94	81.69	79.38	88.25	84.22	86.41	**90.16**
7	35.34	54.11	61.96	64.24	71.13	58.18	76.12	69.63	72.34	**78.68**
8	32.53	40.40	44.53	44.07	48.21	45.25	52.85	48.22	44.58	**52.96**
9	**73.86**	56.82	62.52	61.44	69.11	56.20	71.51	65.00	71.38	70.32
10	31.22	57.91	60.64	59.66	63.96	63.02	65.05	61.78	63.02	**69.50**
11	43.42	58.33	63.72	61.07	68.39	64.04	71.05	71.13	65.73	**74.60**
12	25.25	49.36	71.58	68.36	70.27	**73.82**	67.26	72.18	68.46	68.60
13	25.41	39.09	89.91	88.90	**95.04**	47.16	92.44	91.27	91.38	90.47
14	87.99	88.04	87.26	90.14	94.14	94.92	93.97	91.82	90.05	**99.55**
15	91.66	92.11	97.28	94.72	97.97	96.08	**99.91**	98.87	97.73	99.54
OA	59.06 ± 3.09	68.82 ± 3.62	76.17 ± 2.31	74.98 ± 2.36	78.73 ± 1.67	74.92 ± 1.85	80.76 ± 1.65	78.80 ± 1.70	78.21 ± 1.76	**81.82 ± 2.31**
AA	62.16 ± 1.92	69.70 ± 2.94	78.72 ± 1.76	77.62 ± 2.27	81.28 ± 1.48	76.19 ± 1.76	83.27 ± 1.72	81.32 ± 1.44	80.80 ± 1.49	**84.43 ± 2.01**
$\kappa \times 100$	55.93 ± 3.23	66.32 ± 3.85	74.26 ± 2.50	72.98 ± 2.55	77.03 ± 1.80	72.89 ± 2.00	79.21 ± 1.79	77.10 ± 1.84	76.47 ± 1.89	**80.37 ± 2.50**

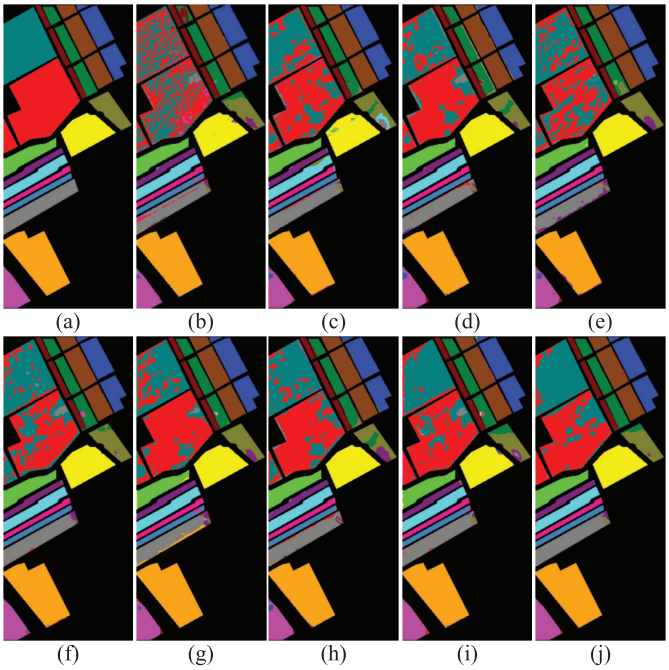


Fig. 16. Classification maps of each method on the SA dataset. (a) Ground truth. (b) 3DCNN (83.49%). (c) DCFSL (89.59%). (d) Gia-CFSL (89.27%). (e) CMFSL (89.54%). (f) ADAFSL (91.17%). (g) DMCM (91.39%). (h) HyMuT (91.13%). (i) GCC-FSL (90.46%). (j) PMC-CFR (92.78%).

challenging Bare Soil (class 6) and Self-Blocking Bricks (class 8) classes, respectively. Fig. 14 presents the classification maps of each method. PMC-CFR performs well in the edge areas of Gravel (class 3) and Bitumen (class 7), effectively reducing the dependence on adjacent terrain in HSIC, owing to the neighboring-pixel removal mechanism during feature extraction.

3) Performance on the Houston Dataset: Table V lists the classification results on the Houston dataset. Compared to the second-best method, PMC-CFR achieves improvements of 1.06%, 1.16%, and 1.16% in OA, AA, and κ ($\times$100), respectively, maintaining its superiority. The accuracy for most classes remains the highest, with the accuracy for Tennis court (class 14) class even improving to 99.55%. Fig. 15 visualizes the classification maps of different methods. The proposed method can accurately classify even highly dispersed small-scale objects, which demonstrates the effectiveness of refining features from different levels after suppressing the impact of potential feature redundancy based on SCL.

4) Performance on the SA Dataset: The results for the SA dataset are shown in Table VI. PMC-CFR still demonstrates strong generalization ability, achieving the best results with OA, AA, and κ ($\times$100) values 1.39%, 0.37%, and 1.54% higher than those of the second-best method, respectively. The classification performance across all classes is also satisfactory, with a 2.75% improvement for the challenging Grapes_untrained (class 8) class. The classification maps of different methods are shown in Fig. 16. Compared to other methods, PMC-CFR has fewer noise points across the regions, possibly benefiting from utilizing contrastive information to reduce the noise interference caused by feature redundancy.

The TD selects 1–5 labeled samples per class for training to explore the impact of different numbers of labeled samples on classification performance. Fig. 17 shows the corresponding OA of various deep learning-based methods. It can be observed that the classification results of all methods improve with the increasing number of labeled samples, and the performance of 3DCNN is generally inferior to that of FSL-based methods. Compared to other methods, PMC-CFR consistently achieves the best classification results under various numbers of labeled samples, with a prominent superiority on the IP dataset. Even in cases with extremely limited labeled samples, the proposed method still demonstrates excellent generalization ability.

F. Ablation Study

The key to PMC-CFR is PMC, whose generated positive pairs drive the MLP to coordinate cross-stage feature

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TABLE VI

CLASSIFICATION RESULTS (%) OF EACH METHOD ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Class	SVM	3DCNN	DCFSL	Gia-CFSL	CMFSL	ADAFSL	DMCM	HyMuT	GCC-FSL	PMC-CFR
1	97.07	96.04	99.15	98.46	**99.71**	99.16	99.12	99.23	99.21	99.56
2	95.32	97.85	99.45	99.41	99.80	99.80	99.95	99.87	99.20	**99.98**
3	72.20	88.35	90.03	90.09	90.02	**97.85**	94.82	90.09	89.94	94.34
4	93.18	97.92	99.52	99.20	99.32	98.20	99.46	99.45	99.57	**99.70**
5	94.96	90.07	89.67	89.91	94.38	96.24	94.99	90.33	89.73	**96.61**
6	90.92	98.73	99.31	99.38	**99.48**	98.96	99.37	98.62	99.27	99.36
7	97.06	98.36	99.43	99.19	**99.50**	99.03	98.53	98.41	99.41	99.17
8	48.11	64.30	78.12	79.02	77.40	79.39	77.96	80.57	82.57	**85.32**
9	96.34	97.34	99.65	98.00	99.55	99.16	**99.79**	99.53	99.19	99.75
10	75.22	78.94	89.88	83.75	86.60	90.37	**92.33**	90.04	86.92	91.44
11	87.95	94.64	98.43	96.47	98.44	97.83	98.92	97.22	95.68	**99.20**
12	93.84	98.92	99.55	98.21	98.87	99.45	**99.81**	99.73	98.41	98.31
13	97.83	97.98	99.26	98.95	98.81	99.53	**99.91**	99.74	98.77	99.46
14	88.44	91.78	**97.84**	97.05	97.70	97.77	97.39	97.36	97.16	97.18
15	65.05	60.57	71.33	72.95	71.84	76.95	**79.73**	79.42	77.59	78.61
16	86.03	88.53	93.92	93.23	93.74	95.47	97.19	94.57	93.20	**97.20**
OA	78.65 ± 2.82	83.49 ± 2.21	89.59 ± 1.65	89.27 ± 1.40	89.54 ± 1.40	91.17 ± 0.70	91.39 ± 1.34	91.13 ± 1.12	90.96 ± 0.87	**92.78 ± 1.04**
AA	86.22 ± 2.22	90.02 ± 1.31	94.04 ± 0.92	93.33 ± 1.23	94.07 ± 1.01	95.32 ± 0.64	95.58 ± 0.59	94.64 ± 1.02	94.11 ± 0.89	**95.95 ± 0.94**
$\kappa \times 100$	76.42 ± 3.04	81.67 ± 2.43	88.42 ± 1.83	88.07 ± 1.56	88.37 ± 1.55	90.17 ± 0.78	90.43 ± 1.48	90.14 ± 1.24	89.95 ± 0.96	**91.97 ± 1.17**

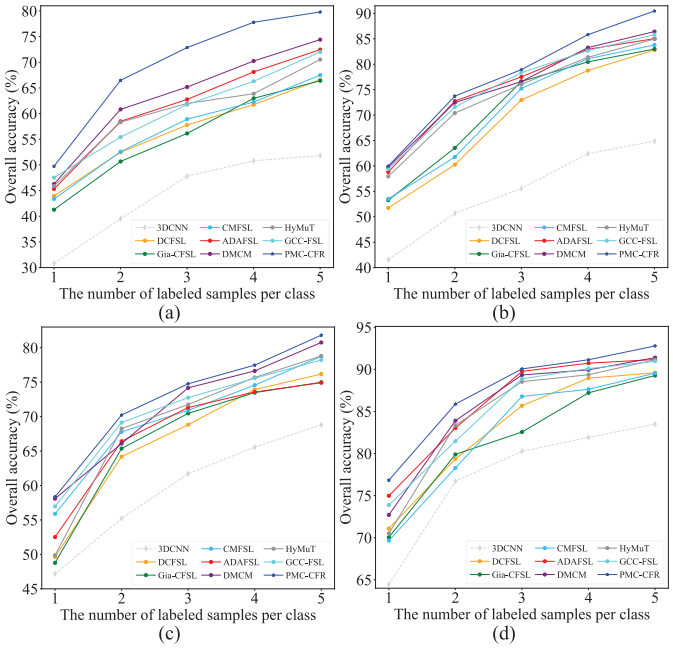


Fig. 17. OA (%) under different numbers of labeled samples per class on the target datasets. (a) IP. (b) UP. (c) Houston. (d) SA.

TABLE VII

ABLATION COMPARISON OF EACH VARIANT OF PMC-CFR

Module			Dataset			
PMC	MLP	DC	IP	UP	Houston	SA
×	×	×	68.49	83.34	77.65	88.47
✓	×	×	71.32	85.95	79.96	90.83
×	×	✓	66.71	82.26	74.99	90.49
✓	✓	×	73.58	86.27	80.37	91.29
✓	×	✓	76.09	88.97	81.44	91.74
✓	✓	✓	**79.79**	**90.48**	**81.82**	**92.78**

refinement, making PMC a prerequisite for cross-stage processing. Table VII lists the OAof different ablation variants. The baseline model consists only of NRMFF trained under the meta-learning paradigm. For the IP, UP, and Houston datasets, the performance improves after introducing PMC but decreases when only DC is introduced. This is because DC aims to cluster features within classes and separate features between classes, and its effectiveness directly depends on the discriminability of the features. For these three datasets, more complex feature extraction is required to achieve strong class distinction. Directly applying DC causes the features to overfit the very limited training samples, failing to generalize well to the test samples. In this case, DC reduces the perturbation among features, resulting in a negative impact. When both PMC and DC are introduced, the classification performance improves significantly, as PMC suppresses the interference of feature redundancy, thereby facilitating the generation of high-quality features, while the refinement of DC further aids in uncovering more discriminative prototypes. For the SA dataset, the classification performance is generally better, and class features are less prone to confusion, allowing DC alone to effectively enhance classification. Introducing MLP-based feature refinement for each dataset is also beneficial. It effectively leverages intradomain positive pairs to guide the model in focusing on key features within each domain while disregarding potential redundancy, thereby facilitating the PMC in alleviating the impact of redundant negative transfer. Meanwhile, the two-stage training it introduces enables the coordination of MLP and DC for cross-stage feature refinement, further amplifying the constraint effect of DC to exploit the classification potential of features at different levels to increase classification performance. Overall, the baseline model does not explicitly suppress feature redundancy, limiting its cross-domain generalization ability. However, by leveraging the potential redundancy of similar prototype components as guidance, PMC is conducted, subsequently driving cross-stage feature refinement, which gradually leads to significant performance improvements. This demonstrates that the collaborative effect of different modules effectively mitigates the negative transfer impact of intradomain feature redundancy on cross-domain learning.

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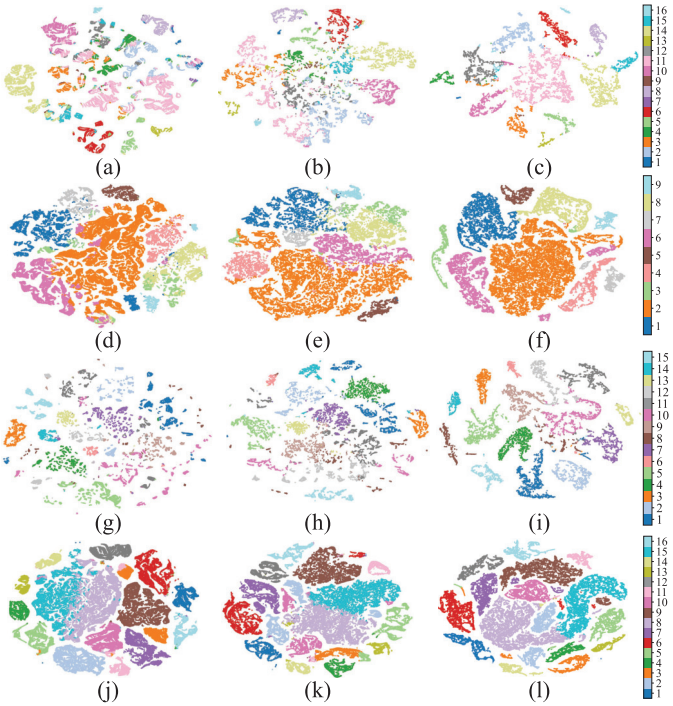


Fig. 18. 2-D feature visualization on four datasets. (a) DMCM for IP. (b) GCC-FSL for IP. (c) PMC-CFR for IP. (d) DMCM for UP. (e) GCC-FSL for UP. (f) PMC-CFR for UP. (g) DMCM for Houston. (h) GCC-FSL for Houston. (i) PMC-CFR for Houston. (j) DMCM for SA. (k) GCC-FSL for SA. (l) PMC-CFR for SA.

G. Feature Visualization of Different Methods

To evaluate the predictive performance of features from an intuitive visual perspective, we perform 2-D visualization of the high-dimensional features extracted by DMCM, GCC-FSL, and PMC-CFR on the four datasets using t-SNE, as shown in Fig. 18. DMCM is the best-performing contrastive method across the four datasets, while GCC-FSL achieves excellent performance enabled by SCL. Clearly, in Soybean-mintill (class 11) of the IP dataset, the contrastive methods exhibit excessive dispersion. Similarly, in the Houston dataset, both DMCM and GCC-FSL tend to produce scattered sample distributions across multiple classes, such as Commercial (class 8), Highway (class 10), and Parking lot 1 (class 12). However, the feature patterns captured by PMC-CFR exhibit clear classification boundaries and more compact interclass clustering. In the UP dataset, the Asphalt (class 1) class exhibits dispersed clustering or excessive overlap in the contrastive methods, whereas the proposed method avoids these issues. In the SA dataset, PMC-CFR achieves the least feature entanglement in Grapes_untrained (class 8) and Vinyard_untrained (class 15). In summary, PMC-CFR demonstrates better interclass separation and intraclass clustering across all datasets.

H. Visualization Comparison of Cross-Domain Performance

To further validate the effectiveness of the proposed method in cross-domain FSL, we conducted t-SNE visualization of

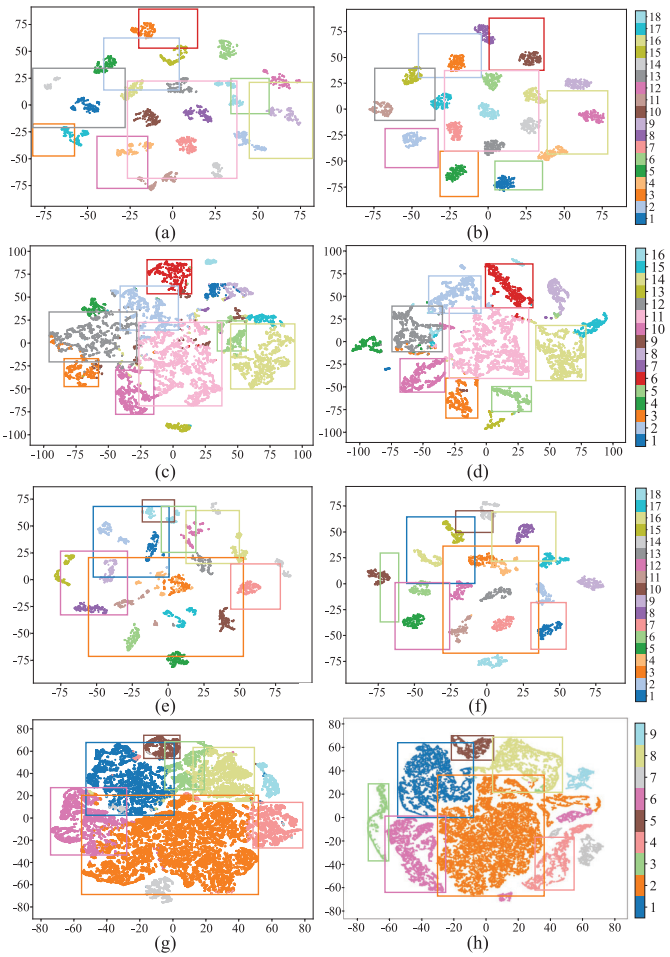


Fig. 19. Feature visualizations of baseline model (left column) and nonablated PMC-CFR (right column) on SD (Chikusei) and TD (IP and UP). (a) Baseline and (b) PMC-CFR for Chikusei (TD: IP). (c) Baseline and (d) PMC-CFR for IP (SD: Chikusei). (e) Baseline and (f) PMC-CFR for Chikusei (TD: UP). (g) Baseline and (h) PMC-CFR for UP (SD: Chikusei).

feature distributions in the SD and TD on IP and UP, comparing the distribution differences between two cases: the baseline model without mitigating the impact of feature redundancy and the nonablated PMC-CFR with mitigating the impact of feature redundancy, as shown in Fig. 19. Although DC can be used independently, it compromises the generalization of features extracted by the baseline model on IP and UP, and is, therefore, not included in the compared baseline model. Within the SD or TD, mitigating the impact of redundancy enables the model to focus more on discriminative features, leading to more compact feature distributions, reduced interclass confusion, and clearer decision boundaries. In cross-domain feature distributions, we observe that when redundancy is not mitigated, certain TD classes may align with multiple SD classes. However, after mitigating redundancy, these classes tend to align more closely with a single SD class, such as Corn-notill (class 2) and Soybean-notill (class 10) in IP and Gravel (class 3) in UP. Although some TD classes may still align with multiple SD classes, their alignment range is significantly reduced, as seen in Soybean-mintill

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TABLE VIII
FLOPS, TRAINABLE PARAMETERS, TRAINING TIME, AND TESTING TIME OF VARIOUS FSL-BASED METHODS ON FOUR DATASETS

Datasets	Metrics	DCFSL	Gia-CFSL	CMFSL	ADAFSL	DMCM	HyMuT	GCC-FSL	PMC-CFR
IP	FLOPs (M)	47.36	66.25	28.38	34.97	9.49	98.42	43.29	15.39
	Params (M)	4.26	0.30	0.63	4.25	0.22	4.69	0.24	0.21
	Training time (s)	3038.12	6497.51	1752.08	3098.05	4362.39	6353.99	6679.53	3814.35
	Testing time (s)	1.63	1.57	1.08	1.43	1.48	1.52	1.49	1.10
UP	FLOPs (M)	46.57	129.53	28.00	34.19	8.70	96.84	42.50	14.63
	Params (M)	4.25	0.84	0.62	4.24	0.21	4.67	0.23	0.21
	Training time (s)	1847.46	4101.44	1045.57	1832.91	2519.13	3863.43	4204.85	1926.70
	Testing time (s)	5.84	5.76	4.01	5.12	5.28	5.78	5.74	4.09
Houston	FLOPs (M)	46.91	130.28	28.33	34.52	9.04	97.51	42.84	14.95
	Params (M)	4.25	0.85	0.63	4.24	0.22	4.68	0.23	0.21
	Training time (s)	2788.44	5766.09	1519.98	2880.04	3873.12	5778.42	5899.26	3479.20
	Testing time (s)	2.15	2.13	1.37	1.95	2.05	2.10	2.08	1.45
SA	FLOPs (M)	47.39	130.86	28.82	35.01	9.52	98.48	43.32	15.42
	Params (M)	4.26	0.85	0.63	4.25	0.22	4.69	0.24	0.22
	Training time (s)	3086.16	6479.93	1963.51	3291.47	4483.24	6319.13	6753.23	3437.11
	Testing time (s)	7.88	7.82	5.42	7.15	7.51	7.89	7.92	5.83

(class 11) and Woods (class 14) of IP, and Asphalt (class 10) and Meadows (class 2) of UP. This phenomenon may result from the unique characteristics of these SD classes, which are beneficial for the corresponding TD classes to learn. Additionally, some TD classes remain difficult to align with the SD due to differences in the class label space. Compared with the baseline model that does not consider feature redundancy, the proposed PMC-CFR first mitigates intradomain redundancy dependence through the PMC, and then focuses on domain-specific key features via the shared MLP and refines multilevel features in conjunction with DC. These observations indicate that the tight collaboration among all modules contributes to enhancing feature discriminability and cross-domain consistency, suppressing the interference of intradomain redundant information in cross-domain FSL, and reducing unnecessary negative transfer caused by redundant knowledge. This alleviates the issue of ambiguous TD class alignment arising from learning redundant features from multiple SD classes, thereby promoting cross-domain feature distribution alignment.

I. Computational Complexity

As shown in Table VIII, we compare the computational complexity of each FSL-based method using the numbers of floating-point operations (FLOPs), trainable parameters, training time, and testing time. PMC-CFR has higher computational efficiency than most of the other methods. It is noteworthy that the proposed method extracts features with stronger discriminative ability through additional operations of mask contrast and feature refinement, thereby effectively reducing FLOPs and model parameters. Although PMC-CFR requires more training time compared to methods like DCFSL and CMFSL, it achieves better classification performance with fewer FLOPs and trainable model parameters.

IV. CONCLUSION

This article proposes the PMC-CFR method for cross-domain FSL in HSIC, which explores potential redundancy from the same feature channel in intradomain prototypes as contrastive information, thereby driving comprehensive feature refinement across different levels. Specifically, NRMFF extracts discriminative features through a neighboring-pixel removal mechanism and multiple convolutional strategies. Next, PAMG identifies potential redundancy based on the statistical distribution of intradomain prototypes to generate contrastive pairs, which are further processed through GC for SCL. The PMC evaluates feature importance in class representation to optimize the dynamic triplet loss, enabling more effective semantic discrimination. In addition, under the iterative two-stage training mode, the shared MLP, driven by positive pairs, performs hierarchical feature modeling by focusing on key components in each domain, while the DC refines interclass and intraclass features across levels by balancing direction and distance to facilitate semantic separation. Their synergy drives cross-stage feature refinement for acquiring more fine-grained representations. Experimental results on TD datasets indicate that the proposed PMC-CFR outperforms many advanced contrastive methods, and feature visualizations further support its effectiveness in cross-domain feature distribution alignment.

Although the proposed method treats similar features among intradomain prototypes as potential redundancy, effectively alleviating their negative impact, feature redundancy is inherently a challenging concept to define. We are unable to fully explore the true specific manifestations of intradomain feature redundancy and its influence under cross-domain interactions. In future work, we will focus on addressing these issues.

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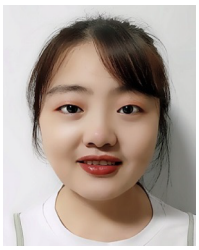
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